Commons Category Suggester: Progress

Megathon M2 · Oct 11, 2026

1. Current approach and changes since M1

The core is unchanged: one open CLIP model (ViT-B/32) embeds images, titles, descriptions and category names into one space, and five scorers rank candidates (similar images, image vs name, similar titles, title vs name, Commons search). Changes since M1:

- **Learned fusion replaces fixed weights:** a logistic regression trained on validation data scores each candidate from 17 features (each scorer's score, log-score and rank; category frequency; found only by search).
- **Tuning objective fixed:** tuning for recall alone pushed obscure search hits to rank 1 (top-1 fell to 0.37); we now tune for top-1 accuracy plus top-10 recall.
- **New tooling:** robustness and speed benchmarks, and a local demo showing each suggestion with the similar images behind it and accept/reject buttons; accepted labels join the index instantly.
- **Reliability:** API calls retry on errors; every step resumes from its cache.

2. Experiments and results

Learned fusion is our best system: the top suggestion is correct for **71%** of new uploads (M1: 65%; image only: 43%). Data: 5,966 images from 43 seed categories; test set = the 1,191 newest uploads.

Method	P@1	R@5	R@10	R@10, non-seed labels
Image only	0.427	0.366	0.406	0.026
+ title and description	0.662	0.496	0.513	0.030
+ Commons search, fixed weights (M1)	0.647	0.550	0.576	0.188
+ Commons search, learned fusion	**0.713**	**0.560**	**0.590**	**0.208**

- **By category type** (top-1, image only → learned fusion): scene 0.61 → 0.85, event 0.41 → 0.73, abstract 0.47 → 0.68, object 0.28 → 0.62. Text helps events and abstract ideas most, as M1 predicted.
- **Search reaches unseen categories:** recall on labels outside the 43 seeds rose from 0.03 to 0.21, but search finds only 22% of them at all.

- **Learned fusion beats every single scorer** (best alone: similar titles, 0.685); the fixed weights did not.

3. Progress on testing

- **Leak checks (done):** temporal 70/10/20 split, near-duplicates removed, everything fit on validation only. Uploaders unseen in training score the same (P@1 0.663 vs 0.653), so photo series do not inflate results.
- **Robustness (done):** degraded test images scored against the clean index, with the M1 fixed weights:

Test images degraded to	P@1, image only	P@1, all scorers	R@10, all scorers
Clean	0.419	0.647	0.576
Blur (radius 3)	0.389	0.666	0.560
JPEG quality 10	0.395	0.648	0.570
48 px low resolution	0.403	0.669	0.568

Image-only top-1 drops 2–3 points; the full system loses at most 1.6 points of recall, since the title and search ignore pixels.

- **Speed and memory (done):** about 35 ms of CLIP encoding and 2 ms of scoring per upload on a laptop GPU; the Commons search call (about 820 ms) dominates. Batched: 113 images/s (71 on CPU). Index: 4 KB per image, about 13 GB for 100 million images with product quantization.
- **To do:** hand review of "wrong" suggestions, and a breakdown by region and language.

4. Plan for the remaining time

1. **Broader data (in progress):** 6,000 random Commons images, reported separately from seed images.
2. **Tune defaults:** neighbour count, temperatures and the search filter, on validation data.
3. **Better name matching:** clean category names and average several text prompts.
4. **Better search queries:** phrases and description words, to raise the 22% ceiling.
5. **Hierarchy pass:** keep the most specific category; partial credit for parents.
6. **Finish testing and demo:** hand review, region audit, learned fusion in the demo, presentation walkthrough.